

GOVT 6019: Homework 2 — Describing Data

Fall 2026

Melissa Sands (Instructor)

Izza Malik (Teaching Assistant)

Part A — The algebra of center and spread

Question 1: The balancing point

- a. Show that for *any* numbers $X_1, \dots, X_n$,

$$\sum_{i=1}^n (X_i - \bar{X}) = 0.$$

- b. Now generalize. Show that for any constant c ,

$$\sum_{i=1}^n (X_i - c) = n(\bar{X} - c),$$

and conclude that the deviations from c sum to zero *if and only if* $c = \bar{X}$.

- c. In one or two sentences: why does (a) disqualify the “average deviation” $\frac{1}{n} \sum_i (X_i - \bar{X})$ as a measure of spread, and what are the two repairs lecture offered?

Question 2: Rescaling without fear

Let $Y_i = a + bX_i$ for every observation, where a and b are constants — a **linear transformation**. Note that you’ve seen this before, e.g., when we recoded survey responses from a 1–4 agreement item into a 0–1 importance score.

- a. Show that $\bar{Y} = a + b\bar{X}$.
- b. Show that $s_Y^2 = b^2 s_X^2$, and therefore $s_Y = |b| s_X$. Then answer two short follow-ups: why did a disappear entirely, and why does the SD formula need the absolute value?
- c. Write the rescale $(4 - x)/3$ in the form $a + bx$, identifying a and b . Then, in R: compute the mean and SD of `important_gayr` *after* recoding the Don’t knows to NA; use your formulas from (a) and (b) to **predict** the mean and SD of the rescaled 0–1 version; and finally rescale in R to confirm your prediction to several decimal places.

```
library(tidyverse)
voter <- read_csv("data/voter2012.csv") # adjust to your project's layout

gayr <- na_if(voter$important_gayr, 5) # 5 = Don't know, not data
mean(gayr, na.rm = TRUE)
sd(gayr, na.rm = TRUE)
# now rescale to 0-1 and check your predictions
```

- d. A colleague glances at your output and concludes: “the rescaled item’s SD is only a third of the original’s — rescaling made the measure less noisy.” In one or two sentences, what is wrong with this?

Question 3: The mean minimizes squared deviations

Suppose you must summarize $X_1, \dots, X_n$ with a single number c , and you count the cost of the summary as the total squared distance from the data:

$$g(c) = \sum_{i=1}^n (X_i - c)^2.$$

You will show that the cost-minimizing choice is exactly $\bar{X}$.

- Add and subtract $\bar{X}$ inside the square, $X_i - c = (X_i - \bar{X}) + (\bar{X} - c)$, and expand $g(c)$ into three sums.
- One of the three terms is always zero. Say which, and why.
- Conclude that

$$g(c) = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - c)^2,$$

and explain why this means g is minimized at $c = \bar{X}$ and nowhere else. What is the minimum value of g , and how is it related to the sample variance s^2 ?

- Parts (a) to (c) need no calculus. If you know calculus, work the derivative version too; it is in the solutions.

Part B — Measurement

Question 4: Reliability or validity?

For each vignette below: (1) diagnose the *primary* problem as one of **reliability** or **validity**; (2) be specific — which kind of reliability or validity is at stake; and (3) propose a one-sentence improvement.

- Two research assistants code the same 100 newspaper-reported protest events using the same written codebook. Asked to classify each event as violent or nonviolent, they agree on only 61 of the 100.

- b. Survey respondents are asked whether they voted in November; asked again three months later, almost all give the same answer they gave the first time. But matched against official records, the survey turnout rate is 14 points higher than the records show.
- c. A researcher measures *political interest* with a ten-question factual quiz (naming officeholders, institutional rules). Scores are highly stable on re-testing. A critic objects that the quiz mostly captures schooling and news exposure, not how much politics interests anyone.
- d. A cross-national project codes a country as democratic if and only if it held a multiparty election in the last five years. Coders almost never disagree. Critics point to countries that hold regular multiparty elections while jailing opposition leaders and controlling the press.

Question 5: Defend or attack this index

A colleague proposes measuring **political engagement** for every VOTER respondent as an additive index of three ingredients, using the raw codes exactly as they sit in `voter2012.csv`:

$$\text{engagement}_i = \frac{\text{pol_interest}_i + \text{educ}_i + \text{important_econ}_i}{3}.$$

Write a short memo (aim for 200 words or fewer) to this colleague. Identify **at least three distinct problems** with the index as proposed, and label each one *fixable* (say how) or *fatal* (say why). End with a verdict: repair it, or abandon it? If repair, list the exact steps in order; if abandon, say what, if anything, in this file could measure engagement more defensibly.

Part C — Applied: the social-issues importance index

In lab you built the four-item importance index, and the closing exercises built a leaner two-item version from the economic items (the economy and the budget deficit). That leaves the social pair — `important_gayr` and `unimportant_abort` — which you have not yet cleaned yourself. Keep the codebook open; one of these two items is reverse-framed.

```
library(tidyverse)
voter <- read_csv("data/voter2012.csv") # adjust to your project's layout
```

Question 6: Build it

- a. Recode the Don't knows (code 5) on both items to `NA` — into **new columns**, leaving the raw ones untouched. How many 5s did each item contain? Audit the recode: report `sum(is.na())` for each item before and after, and confirm the counts add up.
- b. Flip the reverse-framed item. Audit the flip with a crosstab of raw against flipped values, and say where the 2,432 respondents who *strongly disagreed* that abortion is unimportant ended up.

- c. Rescale both cleaned items to 0–1 with higher = more important, then combine them into a two-item index with `rowMeans(..., na.rm = TRUE)`. Report: how many of the 8,000 respondents receive a score, and the index’s mean, median, and SD.

```
# one route to the rowMeans step, once gayr_01 and abort_01 exist:
voter$social_index <- voter |>
  select(gayr_01, abort_01) |>
  rowMeans(na.rm = TRUE)
voter$social_index[is.nan(voter$social_index)] <- NA # answered neither
```

- d. The `na.rm = TRUE` choice scores anyone who answered at least one item. How many respondents are scored under this rule but would be dropped by listwise deletion (requiring both items)? How many receive no score under either rule?
- e. Compare the index’s mean and median to each other — and then to the lab’s *economic* pair (mean 0.90, median 1.00, jammed against the ceiling). What does the comparison say about the two distributions’ shapes? Confirm with a quick histogram (`binwidth = 1/6` works well for a two-item index that lives on sixths).

Question 7: Don’t know, but louder

How much damage do disguised nonresponse codes actually do? Use the `raw important_gayr` (scale 1–4, plus 5 = Don’t know) and compare three treatments of the Don’t knows:

- **Rule 1 (careless):** keep the 5s as data;
- **Rule 2 (missing-data rule):** recode 5 to NA;
- **Rule 3 (the tempting shortcut):** recode 5 to the scale midpoint, 2.5 — “they probably feel average.”

```
x_keep <- voter$important_gayr
x_na <- na_if(voter$important_gayr, 5)
x_mid <- if_else(voter$important_gayr == 5, 2.5, voter$important_gayr)
```

- a. How many respondents chose Don’t know, and what share of the item’s nonmissing answers is that?
- b. Compute the mean and the median under each of the three rules; arrange the six numbers in a small table.
- c. In two or three sentences: which summary moved and which never budged, and why?
- d. (**Stretch.**) The careless mean is a weighted mean in disguise: if a share p of answers are Don’t knows and the honest mean of the rest is m , then keeping the 5s gives exactly $(1 - p)m + 5p$. Verify this numerically. Then use the formula twice: predict the careless mean for a hypothetical item with the *same* honest mean but $p = 0.20$; and explain why the same 39 Don’t knows distort `important_econ` (honest mean about 1.12) more than they distort `important_gayr` (honest mean about 2.68).

Question 8: One good graph

Make **one presentation-quality figure** comparing the distribution of your Question 6 index across the three party groups (`pid3`, as a labeled factor). Any sensible geometry is fine — grouped boxplots, faceted histograms, bars of shares — but it must clear the bar you would set for a figure in a paper:

- factor levels labeled with words, never raw codes;
- `theme_gov()` and the course palette;
- a title that states the figure’s message as a sentence;
- axes labeled, with units or scale made clear;
- if you use bars, they start at zero;
- and a **two-sentence caption**: first sentence, the substantive finding; second sentence, a caveat or scope note (remember whose data this is and what year it comes from).

One thing to watch: the party groups differ a lot in size, so raw counts are not comparable across groups — compare *shares* within each group instead.

If you want more: Gailmard, sections 2.2–2.3 covers the central-tendency and dispersion material in Part A at exactly this level; Healy, *Data Visualization: A Practical Introduction*, ch. 1, is the best short read before attempting Question 8; and Ansolabehere, Rodden & Snyder, “The Strength of Issues,” *APSR* 102(2), 2008, is the measurement argument that justifies ever building an index at all.